

Some notable institutes for instrumentation engineering

- IITs
- Netaji Subhas Institute of Technology, New Delhi
- NITs
- Madras Institute of Technology, Anna University, Chennai
- Birla Institute of Technology & Science
- JSS Academy of Technical Education, Bengaluru
- PES College of Engineering, Mandayalndian Institute of Science, Bengaluru

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to express project requirements for the design and development of hardware suitable for the task. As instrumentation engineers usually find themselves called upon to solve uniquely challenging problems, they must possess essentials such as excellent problem-solving skills and an ability to think outside the box.

Those who are sound in fundamentals and have specific knowledge of measurement systems will find multiple career options within this industry.

Talking about technical skills, instrumentation engineers typically advance their skills in specific disciplines such as distributed control systems, programmable logic controllers, supervisory control and data acquisition (SCADA), and advanced process control. The use of these disciplines will depend on the exact nature of individual job roles.

Is the money worth it?

According to PayScale.com, an instrumentation engineer earns an average salary of ₹ 362,664 per year. The amount of experience strongly influences income for this job. The average starting salary for instrumen-

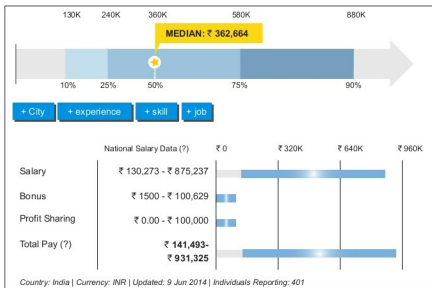


Fig. 4: Approximate pay range of instrumentation engineers (Data courtesy: Payscale.com)

tation engineers in India is ₹ 100,000 to ₹ 240,000 per year. Senior engineers can earn in the range of ₹ 350,000 to 600,000 per annum.

The skills that increase salaries for this job the most are engineering design, distributed control systems and instrument control. Most people move on to other jobs if they have more than ten years' experience in this field.

According to TimesJobs.com's statistics, 64% jobs are for fresh engineers who draw between ₹ 200,000 and ₹ 400,000 per annum. And 17% jobs are for junior associates who get anywhere between ₹ 500,000 and ₹ 700,000 per annum, whereas senior-level employees are paid up to or over ₹ 1,000,000 per annum.

Demand areas

As per data provided by TimesJobs.com, Bengaluru tops the list offering 26% of the jobs in instrumentation in India. Other cities offer: Delhi NCR- 15%, Mumbai - 14%, Chennai - 12% and Pune - 11%. Remaining jobs are scattered all over other locations (refer pie chart).

The maximum number of professionals hired for working in instrumentation sector are in manufacturing and engineering industry with 35% share of jobs, which includes a majority of sales and marketing jobs. Next in line are IT

industry with 22% of jobs, production and maintenance industry with 11% and IT hardware with 7%.

Instrumentation engineering is available as B.Tech or B.E. programmes, entry to which is normally through an all-India entrance test or tests organised by state governments. Institutes like BITS Pilani offer instrumentation with electronics as BE (Hons) electronics and instrumentation.

Challenges

Industry experts believe instrumentation gives the pursuer an edge over mechanical or electrical engineers in terms of wider knowledge base. The pay is low initially but it gets better in later stages, once you develop the skills. It is also a highly responsible role as it includes a multitude and variety of tasks. On the other side, health could be affected due to high noise and temperature levels at manufacturing plants. You could also be required to work for longer hours. The jobs in sales, marketing and software are more as compared to core jobs.

There is hardly any area where instrumentation is not required. Hence, there is vast scope in this field. ●

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